Week 11
الأسبوع 11: من - إلى

Subject/:الموضوع Chemistry **Unit Title : عنوان الوحدةR3.1 Proton transfer reactions** **Grade 12 DP (Year 2)**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 **Neutralization reactions (R3.1.7)** تفاعلات التعادل	• Students will apply their knowledge of neutralization to formulate equations for the reactions of acids with metal oxides, metal hydroxides, hydrogencarbonates and carbonates, and identify the parent acid and base of a salt.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.3 Guide processing with explicit steps, chunking, and interactive models. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	• **Practice Sheet 1 – Neutralization equations (Teams)** • **Lesson slides (class Microsoft Teams channel)**
Lesson 2 **الحصة الثانية** **pH curves (R3.1.8)** منحنيات الرقم الهيدروجيني	• Students will interpret the general shape of the pH curve for a monoprotic strong acid–strong base neutralization, including the pH-axis intercept and the equivalence point.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	• **Practice Sheet 2 – pH curves (Teams)** • **Lesson slides (class Microsoft Teams channel)**
Lesson 3 **الحصة الثالثة** **Exam revision: R3.1.7–R3.1.8** مراجعة للامتحان	• Students will apply their knowledge of neutralization equations and pH curves to answer exam-style questions.	3.4 Support transfer using checklists, organizers, and recurring practice. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	• **Exam revision practice questions (Teams)** • **Use the IB data booklet and a calculator**
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Practice Sheets 1–2, the exam revision questions and the mark scheme are for the class Microsoft Teams channel (share the mark scheme when ready). Remind students to bring a calculator and the IB data booklet. Eye protection applies if a titration practical is run. [Add homework or assessment due date]		

Week # 11 الأسبوع#

Technology Integration دمج التكنولوجيا	Class Microsoft Teams channel for lessons, practice sheets and exam revision questions; students access them on their own devices.
Cross-curricular links الروابط بين المواد الدراسية	Biology: acid–base balance and antacids. Environmental systems: neutralizing acidic soil and water (liming). -Mathematics: sketching and interpreting graphs, and mole calculations.
External Assessment Integration دمج التقييمات الخارجية	IB exam-style questions on neutralization equations and pH curves (Paper 1A multiple choice and Paper 2 structured questions), using the data booklet. Links to the Internal Assessment: Tools 1 and 3 (using titration to find concentration).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL: Thinking skills (interpreting graphs, applying knowledge to new questions), Self-management (exam planning and timing with a revision checklist), Communication (precise use of command terms). Wellbeing: practise exam questions in short blocks and review mistakes calmly.
Standards المعايير	• R3.1.7: Acids react with bases in neutralization reactions. Formulate equations for the reactions between acids and metal oxides, metal hydroxides, hydrogencarbonates and carbonates. Identify the parent acid and base of different salts. • R3.1.8: pH curves for neutralization reactions involving strong acids and bases have characteristic shapes and features. Sketch and interpret the general shape of the pH curve, including the intercept with the pH axis and the equivalence point (monoprotic reactions only).
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Tool 1, Structure 1.1: How can the salts formed in neutralization reactions be separated? Reactivity 1.1: Neutralization reactions are exothermic. How can this be explained in terms of bond enthalpies? Reactivity 3.2: How could we classify the reaction that occurs when hydrogen gas is released from the reaction between an acid and a metal? Structure 1.4: Why is the equivalence point sometimes referred to as the stoichiometric point? Tools 1 and 3, Structure 1.3: How can titration be used to calculate the concentration of an acid or base in solution? TOK: How do we decide whether a simplified model, such as the general shape of a pH curve, is adequate for a particular purpose?
Success Criteria معايير النجاح **(Measurable)**	**Must (DOK 1–2):** I can identify the parent acid and base of a salt and write balanced equations for an acid reacting with a metal oxide, metal hydroxide, hydrogencarbonate or carbonate. **Should (DOK 2–3):** I can sketch the general pH curve for a monoprotic strong acid–strong base neutralization and interpret its pH-axis intercept and equivalence point. **Could (DOK 3–4):** I can explain why neutralization is exothermic in terms of bond enthalpies and evaluate how a pH curve can be used to find the concentration of an unknown acid or base.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding التفاهم العالمي Choose an item. المجال Area: Innovation and Infrastructure التطبيق Implementation: Students link neutralization to real-world applications such as antacids and treating acidic soil or water, connecting chemistry to innovation and infrastructure.